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Power System Frequency Regulation via Multiple Energy Storage Resources Considering Building Climate Zone Diversity

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***Abstract*—The increasing penetration of photovoltaic (PV) power generation has introduced more fluctuations into the power system, which challenges frequency security. To efficiently utilize PV power and provide consumer demand-side flexibility for frequency regulation, a PV direct-driven air-conditioning system with energy storage (PACES) is designed to integrate PV generation with multiple energy storage resources under a direct-current building power architecture. PACES is a building-scale consumer energy system, in which the air conditioner functions as a virtual thermal energy storage system (TESS) by leveraging building thermal inertia and is coordinated with the battery energy storage system (BESS). However, BESS and TESS present heterogeneous characteristics, which pose difficulties in coordinating multiple PACES for frequency regulation. Furthermore, building climate zone diversity leads to different TESS regulation potentials, which subsequently impact the coordination control between BESS and TESS. To address these problems, this research provides a climate-aware PACES control framework that embeds building climate zone diversity into coordinated PACES regulation. A unified state-space model incorporating equipment and building climate characteristics is established to represent both BESS and TESS. Based on this model, a distributed consensus control strategy is proposed for multiple PACES. The proposed strategy is validated through RTDS-based hardware-in-the-loop experiments under several scenarios. Under the tested PV-induced power deficit, PACES regulation improves the frequency nadir from 49.769 Hz to 49.925 Hz and reduces the nominal-frequency deficit from 0.231 Hz to 0.075 Hz. Sensitivity analysis over five typical Chinese building climate zones demonstrates the climatic adaptability of the proposed strategy.**

***Index Terms*—PV direct-driven air-conditioning system with energy storage, distributed consensus control, frequency regulation, building climate zone diversity**

NOMENCLATURE

Abbreviation

BESS	Battery energy storage system
DN	Distribution network
DSO	Distribution system operator
HIL	Hardware-in-the-loop
IoT	Internet of things
PACES	Photovoltaic direct-driven air-conditioning system with energy storage
PV	Photovoltaic
RMSE	Root mean square error
RTDS	Real-time digital simulator
SOC	State of charge
TESS	Thermal energy storage system

Parameter

η_{COP}	Coefficient of performance of TESS
Cap_i	Energy capacity of BESS
σ_i	Self-discharge rate of BESS
ζ_i	Indoor-outdoor air exchange rate
I	Total number of PACES
i	Index of PACES
t	System operating time
U_i^{h}	Heat transfer coefficient of the building envelope

Set

$\mathbf{L}$	Laplacian matrix of the PACES communication graph
$\mathcal{G}$	Communication graph among PACES
$\mathcal{N}_i$	Neighbor set of PACES i
a_{ij}	Communication weight between PACES i and PACES j
b_i	Indicator of whether PACES i receives the DSO regulation signal

Variable

$\mathbf{X}(t)$	Aggregated state vector of all PACES
$\mathbf{Y}(t)$	Aggregated output vector of all PACES
$e(t)$	Global regulation error
$P_i^{\mathrm{BESS}}(t)$	BESS power of PACES i
$P_i^{\mathrm{PV}}(t)$	PV power of PACES i
$P_i^{\mathrm{TESS}}(t)$	TESS power of PACES i
$P_{\mathrm{DEM}}(t)$	System-level regulation demand
$P_{\mathrm{GRID}}(t)$	Power supplied by the DN
$P_{\mathrm{LOAD}}(t)$	Load demand
$P_{\mathrm{REG}}(t)$	Aggregated PACES regulation power
$T_i^{\mathrm{a}}(t)$	Indoor temperature
$T_i^{\mathrm{o}}(t)$	Outdoor temperature
$T_i^{\mathrm{set}}(t)$	Temperature setpoint of the air conditioner
$y_i(t)$	Node-level regulation signal of PACES i

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I. Introduction

To mitigate global warming caused by fossil fuel, integrating renewable energy resources into the power system is becoming more and more important [1]. Photovoltaic (PV) power generation, as a renewable energy resource, is gaining a larger share of power generation, particularly in the distribution network (DN). However, PV power generation is influenced by weather conditions [2], light intensity and other environmental factors, which pose a challenge to DN frequency security [3]. To ensure DN operation, traditional generation-side operating reserves are usually scheduled to balance power fluctuations. However, as PV penetration increases, these generation-side regulation resources face increasing pressure because PV-induced power fluctuations become more frequent and uncertain [4]. Frequency is a critical indicator of DN operation [5]. The imbalance between power supply and demand can lead to frequency deviation and threaten DN frequency security [6]. Therefore, it is necessary to explore new flexible resources, especially consumer demand-side resources, that can provide supplementary frequency regulation support for DNs with high PV penetration.

Demand-side flexibility has attracted increasing attention [7]. Recent research indicates that buildings present significant potential as demand-side flexibility resources [8]. Some scholars propose using Internet of Things (IoT) technology to adjust these demand-side loads for frequency regulation, referred to as demand response [9]. The air conditioner is a common consumer electric equipment in buildings and has been demonstrated to be effective in providing frequency regulation support to the DN [10]. From a flexibility perspective, the air conditioner can be modeled as a thermal energy storage system (TESS) by leveraging the inherent thermal inertia of the building envelope and indoor air. Li et al. [11] investigate a PV direct-driven air-conditioning system and evaluate its zero-energy potential by using TESS flexibility to match PV generation in real time. Due to the thermal inertia of buildings, short-term adjustments to TESS operation can provide power support to the DN while ensuring indoor thermal comfort. They further proposed a part-time part-space operation mode and a thermal comfort temperature range control for PV direct-driven air-conditioning system [12]. These studies demonstrate that coordinating TESS operating mode with indoor temperatures can optimize the PV-TESS energy matching and reduce power fluctuations on the DN. However, the TESS is constrained by inherent physical and behavioral constraints. Therefore, coordinated control of PV and TESS may not sufficiently provide flexibility during long-term PV surpluses or rapid power fluctuations. Consequently, it is necessary to introduce additional flexible resources.

The battery energy storage system (BESS) is another essential flexible resource for DN frequency regulation because of its fast response capability. However, it cannot directly reshape building energy demand or utilize thermal inertia. Therefore, relying solely on BESS may be insufficient to leverage demand-side flexibility fully. Some scholars have proposed the PEDF (photovoltaics, energy storage, direct current, flexibility) concept to formulate a building-level PV and energy storage integrated flexible direct current (DC) distribution system [13]. PEDF coordinates PV, energy storage, and demand-side flexibility within a DC architecture with high scalability and adaptability. Based on the PEDF concept, the PV direct-driven air-conditioning system with energy storage (PACES) is developed as a practical implementation solution for PEDF [14]. PACES can be regarded as a building-scale consumer energy system that integrates PV generation, BESS, and air-conditioning flexibility within a DC building power architecture. With appropriate demand response mechanisms, such consumer-side flexibility can be collected and compensated as a flexible regulation capacity for power system services, such as frequency regulation. Recent studies have shown that pricing mechanisms can encourage demand-side flexibility [15]. In addition, household thermal inertia can be considered in demand response while maintaining thermal comfort [16]. Therefore, consumers may be encouraged to participate in PACES coordination through regulation-service compensation, while the indoor-temperature constraints are used to protect thermal comfort. By coupling PV power generation, energy storage, and thermal demands, DC-based PACES significantly enhances energy utilization efficiency. As shown in Fig. 1, the power from thermal and PV stations is delivered to the DN through substations and transformers. The building with PACES hosts an internal DC energy hub. In this hub, the building adopts a loose-coupled bipolar DC architecture [17]. Specifically, two pole-to-neutral buses (i.e., $+375$ V and -375 V [18]) are regulated independently while sharing a common neutral conductor. In addition, selected high-voltage DC loads can be supplied through the pole-to-pole link (e.g., 750 V [19]). This architecture directly integrates PV and energy storage systems with DC features into the DC bus via DC/DC converters [20]. Compared with alternating-current (AC) architectures, it reduces intermediate conversion stages, improving conversion efficiency and enabling fast, high-bandwidth power regulation [21]. In addition, the loose-coupled bipolar structure decouples the control loops of the two poles. As a result, local disturbances (e.g., load transients or converter-side perturbations) are largely confined to the affected pole rather than propagating across the entire DC system, which improves operational robustness under disruptive events [22]. These inherent DC features make PACES particularly suitable for providing fast DN regulation services, such as frequency regulation, which is the focus of this study.

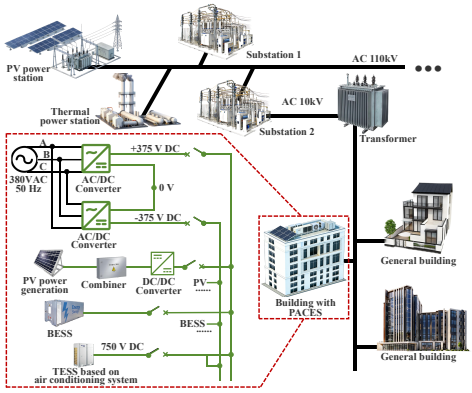


Fig. 1. The structure of the DC-based PACES participates in the frequency regulation.

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However, translating this capability into reliable DN regulation services requires coordinated planning and control of PACES while satisfying DN and comfort constraints. In parallel, several studies investigate coordinated planning and control of PACES. Wan et al. [23] further investigate PACES and employ a hierarchical optimization approach to coordinate BESS regulation capacity design and operating strategies to reduce costs and emissions. Chen et al. [24] investigate a building energy system with PV, BESS, and TESS and optimize the regulation capacities of these demand-side resources. These studies confirm that coordinated use of PV, BESS and TESS can reshape the load profile and reduce energy cost. Nevertheless, they mainly focus on building-level energy management (e.g., energy balance, self-consumption, and sizing) rather than DN frequency regulation. Both BESS and TESS can contribute to DN frequency regulation, but they respond on heterogeneous timescales: BESS provides fast electrical power regulation, whereas TESS reshapes thermal demand more slowly. When these two flexibilities are coordinated within PACES, the system can follow regulation commands and deliver frequency regulation to DNs, enhancing frequency security and enabling higher PV penetration. Hence, coordinating the heterogeneous characteristics of BESS and TESS remains an essential issue for enabling PACES to provide DN frequency regulation.

Distributed control has been widely used to coordinate large numbers of flexible resources [25]. Distributed consensus control enables consistency in the regulation effect of the whole system through information exchange among neighboring nodes [26]. It has been applied to share frequency regulation tasks among flexible resources in a scalable way [27]. Existing studies have demonstrated that TESS can effectively participate in demand response through distributed consensus control [28]. The distributed controllers in these studies can be used to balance the comfort and energy consumption, while maintaining sustainable demand response capabilities in heterogeneous building groups [29]. However, due to the inherent comfort constraints and limited regulation depth of TESS, TESS needs to cooperate with BESS to enhance system reliability. Liu et al. [30] accounted for the heterogeneous dynamic responses of BESS and TESS by establishing a unified state-space model and proposing a distributed coordinated control method. However, the TESS [31] in the unified state-space model is based on a simplified RC model and does not explicitly consider specific building parameters. Therefore, the existing TESS model cannot be directly applied to the BESS-TESS coordination problem of PACES, where the air-conditioning flexibility, battery response, and building-side operating conditions need to be represented in a unified form. Building parameters also play a crucial role in the coordinated control of PACES. Qi et al. [32] demonstrated that climate-induced variations in outdoor environments and building envelope thermal characteristics result in significant regional differences in TESS regulation. According to China's code for thermal design of civil building (GB 50176-2016) [33], the building climate zones are classified into five typical zones: Severe cold zone (SCZ), Cold zone (CZ), Mild zone (MZ), Hot summer and cold winter zone (HSCWZ), and Hot summer and warm winter zone (HSWWZ) [34]. The building climate zone diversity across these regions results in variations in key building parameters (i.e., outdoor temperature, heat transfer coefficient, and indoor–outdoor air exchange rate). However, these parameters have not been explicitly embedded into the TESS model for distributed PACES coordination. This gap makes it difficult to evaluate how building climate zone diversity affects the coordinated regulation capability of BESS and TESS in PACES. Therefore, it is necessary to establish a climate-aware PACES model and control framework that integrates building climate zone characteristics into BESS-TESS coordination.

To bridge the above research gap, this study develops a climate-aware distributed coordination framework for multiple PACES participating in DN frequency regulation. The main contributions of this research are summarized as follows.

- A unified dynamic model is established to describe the heterogeneous characteristics of BESS and TESS in a consistent state-space form. In this model, the air conditioner is represented as TESS through indoor-temperature regulation, and building climate zone diversity is incorporated through thermal parameters. In addition, the disturbance terms caused by PV fluctuations, load variation, and thermal dynamics are included in the PACES modeling framework.
- A distributed consensus control strategy is proposed to coordinate multiple PACES through neighbor-to-neighbor communication. Based on the modeled regulation demand and the aggregated PACES power response, the global regulation error is formed and converted into node-level PACES power references through the consensus update. It can reduce communication dependence and maintain regulation capability under representative communication disturbances.
- RTDS-based hardware-in-the-loop experiments are conducted to validate the proposed modeling and control framework under realistic DN disturbances. The validation covers a 4-node typical topology and a representative 40-node campus topology, together with a centralized droop benchmark, a user behavior uncertainty experiment, a TESS rebound experiment, and communication-robustness experiments.

II. Formulation of PACES for Frequency Regulation

A. Architecture of PACES for Frequency Regulation

Fig. 2 describes the modeling and control framework for PACES participating in power system frequency regulation. The framework includes climate-aware TESS modeling, unified BESS-TESS state-space representation, and distributed coordination among multiple PACES. First, building climate zone diversity is considered by embedding key thermal parameters, including outdoor temperature, heat transfer coefficient, and indoor–outdoor air exchange rate, into the room thermal model, which affects the baseline TESS power and regulation potential. Second, BESS and TESS are represented in a unified state-space form by introducing normalized power and SOC

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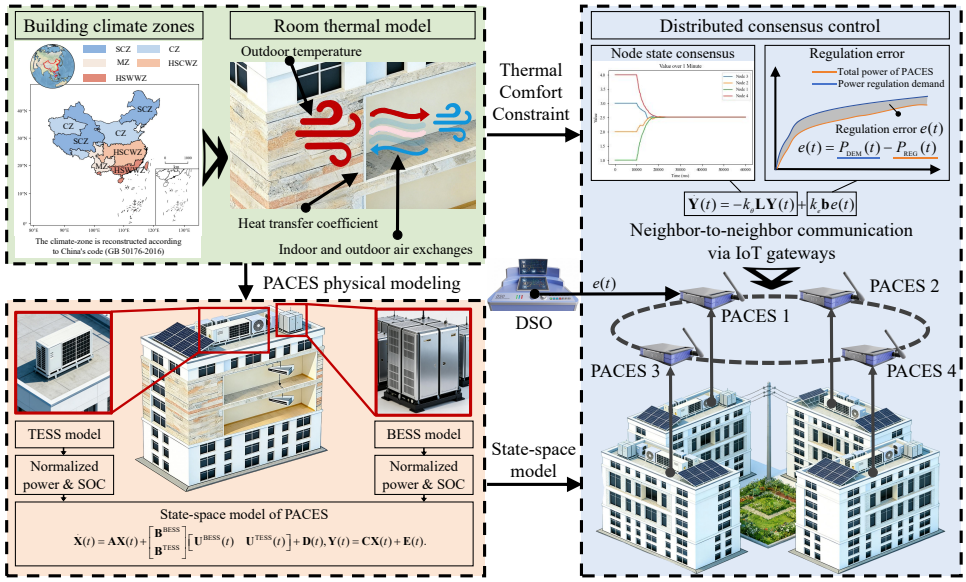


Fig. 2. Overview of modeling and control methodology for PACES participating in power system frequency regulation [35].

variables, so that heterogeneous PACES can be aggregated and coordinated in per-unit quantities. Third, the distribution system operator (DSO) converts the supply-demand imbalance caused by PV power fluctuations into the system-level regulation demand $P_{\mathrm{DEM}}(t)$. The mismatch between $P_{\mathrm{DEM}}(t)$ and the aggregated PACES regulation power $P_{\mathrm{REG}}(t)$ forms the global regulation error $e(t)$ for the distributed consensus controller. Through IoT-based neighbor-to-neighbor communication, this regulation error is converted into node-level PACES power references . In this way, multiple PACES can coordinate BESS and TESS responses while satisfying battery constraints, thermal comfort constraints, and local communication requirements. Accordingly, the following mathematical formulation is organized based on the framework in Fig. 2. The BESS and TESS dynamics are described by the state-space models in Eqs. (12) and (30). Based on these models, the PACES power response is calculated by Eqs. (4)–(37), and the distributed consensus update is given by Eq. (39).

B. Supply-demand Balance for Frequency Regulation

To ensure DN frequency security, the grid power $P_{\mathrm{GRID}}(t)$ should balance the total PACES output $P_i^{\mathrm{PACES}}(t)$ and the load demand $P_{\mathrm{LOAD}}(t)$. Here, i is the PACES index and I is the total number of PACES. The power balance of PACES i can be expressed as follows:

$$P_i^{\mathrm{PACES}}(t) = P_i^{\mathrm{PV}}(t) + P_i^{\mathrm{BESS}}(t) + P_i^{\mathrm{TESS}}(t), \quad (1)$$

where $P_i^{\mathrm{PV}}(t)$ is the PV power of the PACES; $P_i^{\mathrm{BESS}}(t)$ is the BESS power of the PACES; $P_i^{\mathrm{TESS}}(t)$ is the TESS power of the PACES. PV power fluctuations may cause supply-demand imbalance and frequency deviation. Therefore, the regulation demand $P_{\mathrm{DEM}}(t)$ is defined as the power mismatch to be compensated by PACES:

$$P_{\mathrm{DEM}}(t) = P_{\mathrm{GRID}}(t) - P_{\mathrm{LOAD}}(t) - \sum_{i=1}^{I} P_i^{\mathrm{PV}}(t). \quad (2)$$

For each PACES i, the regulation power is provided jointly by the BESS and TESS. The total regulation power $P_{\mathrm{REG}}(t)$ of all PACES can be expressed as follows:

$$P_{\mathrm{REG}}(t) = \sum_{i=1}^{I} P_i^{\mathrm{reg}}(t) = \sum_{i=1}^{I} \left(P_i^{\mathrm{BESS}}(t) + P_i^{\mathrm{TESS}}(t)\right). \quad (3)$$

C. Modelling of BESS

This subsection presents the BESS charging/discharging model, normalized power and SOC, and state-space model. Recent studies on high-precision SOC estimation [36], neural-network-based SOC estimation [37], and electrochemical-thermal coupling modeling of lithium-ion batteries [38] provide useful references for battery state representation and mathematical modeling. These studies provide useful references for BESS modeling and state representation.

1) Charging and Discharging Model: The SOC of the BESS (i.e., $\mathrm{SOC}_{\mathrm{BESS},i}$) is updated based on the power input or output and the BESS's efficiency. The variation of $\mathrm{SOC}_{\mathrm{BESS},i}$ can be expressed as follows:

$$\dot{\mathrm{SOC}}_i^{\mathrm{BESS}}(t) = -\sigma_i \,\mathrm{SOC}_i^{\mathrm{BESS}}(t) + \frac{\eta_i(t)}{\mathrm{Cap}_i} P_i^{\mathrm{BESS}}(t), \quad (4)$$

where σ_i is the self-discharge rate of BESS i; Cap_i is the energy capacity of BESS i; $P_i^{\mathrm{BESS}}(t)$ is the active power of BESS i at time t; $\eta_i(t)$ is the charge and discharge efficiency, which can be expressed as follows:

$$\eta_i(t) = \begin{cases} \eta_i^{\mathrm{in}}, & P_i^{\mathrm{BESS}}(t) \geq 0 (\text{charging state}), \\ \dfrac{1}{\eta_i^{\mathrm{out}}}, & P_i^{\mathrm{BESS}}(t) < 0 (\text{discharging state}), \end{cases} \quad (5)$$

where η_i^{in} and η_i^{out} are the charging and discharging efficiencies, respectively; $P_i^{\mathrm{BESS}}(t) < 0$ indicates that the BESS is in a discharging state while $P_i^{\mathrm{BESS}}(t) \geq 0$ is in charging state. The power of the BESS is constrained by its rated charge and discharge limits:

$$-P_i^{\mathrm{d}} \leq P_i^{\mathrm{BESS}}(t) \leq P_i^{\mathrm{c}}, \quad (6)$$

where P_i^{c} and P_i^{d} are the maximum charging and discharging powers of BESS i, respectively.

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2) Normalized Power and SOC: To obtain a unified representation, the BESS power is normalized by its available regulation capacity. The maximum regulation power of BESS i is defined as follows:

$$P_i^{\mathrm{BESS,max}} = \begin{cases} P_i^{\mathrm{c}}, & P_i^{\mathrm{BESS}}(t) \geq 0(\text{charging state}), \\ P_i^{\mathrm{d}}, & P_i^{\mathrm{BESS}}(t) < 0(\text{discharging state}), \end{cases} \tag{7}$$

where $P_i^{\mathrm{BESS,max}}$ is the magnitude of the maximum charging or discharging power. The normalized BESS power can be written as follows:

$$p_i^{\mathrm{BESS}}(t) = \frac{P_i^{\mathrm{BESS}}(t)}{P_i^{\mathrm{BESS,max}}}, \tag{8}$$

where $p_i^{\mathrm{BESS}}(t)$ is the normalized power of BESS i and satisfies $p_i^{\mathrm{BESS}}(t) \in [-1, 1]$. The normalized SOC of BESS i can be defined based on the admissible SOC range as follows:

$$soc_i^{\mathrm{BESS}}(t) = \frac{\mathrm{SOC}_i^{\mathrm{BESS}}(t) - \mathrm{SOC}_i^{\min}}{\mathrm{SOC}_i^{\max} - \mathrm{SOC}_i^{\min}}, \tag{9}$$

where $soc_i^{\mathrm{BESS}}(t)$ is the normalized SOC; $\mathrm{SOC}_i^{\min}$ and $\mathrm{SOC}_i^{\max}$ are the minimum and maximum allowable SOC levels of BESS i, respectively. The normalized SOC satisfies $soc_i^{\mathrm{BESS}}(t) \in [0, 1]$.

3) State-space Model of BESS: Based on the aforementioned charging/discharging model and the normalized SOC definition, the SOC dynamics of BESS i can be rewritten as follows:

$$\dot{soc}_i{}^{\mathrm{BESS}}(t) = -\sigma_i\, soc_i^{\mathrm{BESS}}(t) + \gamma_i^{\mathrm{BESS}}(t)\, p_i^{\mathrm{BESS}}(t) + \delta_i^{\mathrm{BESS}}, \tag{10}$$

where $\gamma_i^{\mathrm{BESS}}(t)$ is the gain from the normalized power to the SOC and δ_i^{BESS} is a constant term caused by the lower SOC bound and self-discharge. $\gamma_i^{\mathrm{BESS}}(t)$ and δ_i^{BESS} can be expressed as follows:

$$\begin{cases} \gamma_i^{\mathrm{BESS}}(t) = \dfrac{\eta_i(t) P_i^{\mathrm{BESS,max}}}{Cap_i\big(\mathrm{SOC}_i^{\max} - \mathrm{SOC}_i^{\min}\big)}, \\ \delta_i^{\mathrm{BESS}} = -\sigma_i \dfrac{\mathrm{SOC}_i^{\min}}{\mathrm{SOC}_i^{\max} - \mathrm{SOC}_i^{\min}}. \end{cases} \tag{11}$$

The state-space and output model of BESS i can be expressed as follows:

$$\begin{cases} \dot{\mathbf{x}}_i^{\mathrm{BESS}}(t) = \mathbf{A}_i^{\mathrm{BESS}}\, \mathbf{x}_i^{\mathrm{BESS}}(t) + \mathbf{B}_i^{\mathrm{BESS}}\, u_i^{\mathrm{BESS}}(t) + \mathbf{d}_i^{\mathrm{BESS}}(t), \\ y_i^{\mathrm{BESS}}(t) = \mathbf{C}_i^{\mathrm{BESS}}\, \mathbf{x}_i^{\mathrm{BESS}}(t) + E_i^{\mathrm{BESS}}, \end{cases} \tag{12}$$

where

$$\begin{cases} \mathbf{x}_i^{\mathrm{BESS}}(t) = \begin{bmatrix} soc_i^{\mathrm{BESS}}(t) \\ p_i^{\mathrm{BESS}}(t) \end{bmatrix}, \\ \mathbf{A}_i^{\mathrm{BESS}} = \begin{bmatrix} -\sigma_i & \gamma_i^{\mathrm{BESS}}(t) \\ 0 & 0 \end{bmatrix}, \\ \mathbf{B}_i^{\mathrm{BESS}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ \mathbf{C}_i^{\mathrm{BESS}} = \begin{bmatrix} 0 \\ P_i^{\mathrm{BESS,max}} \end{bmatrix}, \\ \mathbf{d}_i^{\mathrm{BESS}} = \begin{bmatrix} \delta_i^{\mathrm{BESS}} \\ 0 \end{bmatrix},\ E_i^{\mathrm{BESS}} = 0, \end{cases} \tag{13}$$

where $\mathbf{x}_i^{\mathrm{BESS}}(t)$ is the state vector; $\mathbf{A}_i^{\mathrm{BESS}}$, $\mathbf{B}_i^{\mathrm{BESS}}$, $\mathbf{C}_i^{\mathrm{BESS}}$ and E_i^{BESS} are the coefficient matrices and variables of the BESS model, respectively; $\mathbf{d}_i^{\mathrm{BESS}}$ is the coefficient term that incorporates the self-discharge effect and the lower limit influence of the SOC.

D. Modelling of TESS

To capture air conditioner flexibility, this section establishes a TESS model based on room thermal dynamics and operational constraints. The model converts indoor-temperature regulation into normalized TESS power and temperature-based SOC, enabling unified BESS-TESS modeling in PACES.

1) Room Thermal Model: The electrical power of TESS is influenced by room thermal dynamics [39], which can be expressed as follows:

$$C_i \frac{dT_i^{\mathrm{a}}(t)}{dt} = H_i^{\mathrm{gain}}(t) - H_i^{\mathrm{TESS}}(t), \tag{14}$$

where

$$C_i = c_i^{\mathrm{a}}\, \rho_i^{\mathrm{a}}\, V_i, \tag{15}$$

where C_i is the building thermal capacity and c_i^{a} is the air heat capacity; ρ_i^{a} is the air density; V_i is the building-scale equivalent indoor air volume served by the air conditioners in PACES i; $T_i^{\mathrm{a}}(t)$ is the indoor temperature at time t; $H_i^{\mathrm{gain}}(t)$ is the heat gain in the room at time t; and $H_i^{\mathrm{TESS}}(t)$ is the cooling produced in the room at time t. The $H_i^{\mathrm{gain}}(t)$ and $H_i^{\mathrm{TESS}}(t)$ can be expressed as follows:

$$\begin{cases} H_i^{\mathrm{gain}}(t) = K_i^{\mathrm{h}}\big(T_i^{\mathrm{o}}(t) - T_i^{\mathrm{a}}(t)\big) + \varepsilon_i A_i, \\ H_i^{\mathrm{TESS}}(t) = \eta_{\mathrm{COP}}\, P_i^{\mathrm{TESS}}(t), \end{cases} \tag{16}$$

where

$$K_i^{\mathrm{h}} = \frac{1}{R_i} + \zeta_i C_i = U_i^{\mathrm{h}} A_i^{\mathrm{S}} + \zeta_i C_i, \tag{17}$$

where R_i is the building thermal resistance; U_i^{h} is the overall heat-transfer coefficient of the building envelope; A_i is the building-scale equivalent floor area served by the air conditioners in PACES i and is used for the internal heat-gain calculation; A_i^{S} is the building-scale equivalent envelope area for outdoor heat exchange and is used for the equivalent heat-transfer calculation; $T_i^{\mathrm{o}}(t)$ is the outdoor temperature at time t; ζ_i is the indoor-outdoor air exchange rate; ε_i is the internal heat gain per unit area caused by occupants and equipment; η_{COP} is the energy efficiency ratio of TESS; $P_i^{\mathrm{TESS}}(t)$ is the active power of TESS i at time t. By simplifying the above equation (i.e., Eqs. (14)–(17)), the temperature model can be obtained as follows:

$$\frac{dT_i^{\mathrm{a}}(t)}{dt} = a_i T_i^{\mathrm{a}}(t) + b_i T_i^{\mathrm{o}}(t) + c_i P_i^{\mathrm{TESS}}(t) + d_i, \tag{18}$$

where

$$a_i = -\frac{K_i^{\mathrm{h}}}{C_i}, b_i = \frac{K_i^{\mathrm{h}}}{C_i}, c_i = -\frac{\eta_{\mathrm{COP}}}{C_i}, d_i = \frac{\varepsilon_i A_i}{C_i}. \tag{19}$$

2) TESS Power Model: The electrical power of the TESS is directly influenced by the compressor operating frequency [40]. The active power of TESS i can be expressed as follows:

$$P_i^{\mathrm{TESS}}(t) = P_i^{\mathrm{TESS}}(t-1) + \kappa_i^{\mathrm{p}}\, \Delta f_i^{\mathrm{TESS}}(t)\big(1 - e^{-\Delta t/\tau_{\mathrm{c},i}}\big) + \mu_i^{\mathrm{p}}, \tag{20}$$

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where κ_i^{p} and μ_i^{p} are the operating power coefficients of TESS; τ_i^{c} is the inertial time constant of TESS's compressor; $\Delta f_i^{\mathrm{TESS}}(t)$ is the TESS's compressor frequency deviation at time t, which can be expressed as follows:

$$\Delta f_i^{\mathrm{TESS}}(t) = f_i^{\mathrm{TESS}}(t) - f_i^{\mathrm{TESS}}(t-1). \tag{21}$$

The compressor frequency of the air conditioner is related to the temperature setpoint T_i^{set} and indoor temperature $T_i^{\mathrm{a}}(t)$, which can be expressed as follows:

$$\begin{cases} \Delta f_i^{\mathrm{TESS}}(t) = \alpha_i \left(T_i^{\mathrm{a}}(t) - T_i^{\mathrm{set}}(t)\right) + \\ \qquad\qquad \beta_i \int \left(T_i^{\mathrm{a}}(t) - T_i^{\mathrm{set}}(t)\right) \mathrm{d}t, \\ f_i^{\mathrm{TESS,min}} \le f_i^{\mathrm{TESS}}(t) \le f_i^{\mathrm{TESS,max}}, \end{cases} \tag{22}$$

where $\alpha_i(T_i^{\mathrm{a}} - T_i^{\mathrm{set}}) + \beta_i \int (T_i^{\mathrm{a}} - T_i^{\mathrm{set}})\mathrm{d}t$ is a proportional-integral (PI) controller; α_i and β_i are the coefficients of PI controllers; and $f_i^{\mathrm{TESS,min}}$ and $f_i^{\mathrm{TESS,max}}$ are the minimum and maximum compressor frequencies of TESS, respectively.

3) Normalized Power and SOC: The baseline power of TESS i is obtained by setting $dT_i^{\mathrm{a}}(t)/dt = 0$ and $T_i^{\mathrm{a}}(t) = T_i^{\mathrm{set}}$ in the room thermal model, which can be expressed as follows:

$$0 = K_i^{\mathrm{h}}\left(T_i^{\mathrm{o}}(t) - T_i^{\mathrm{set}}\right) + \varepsilon_i A_i - \eta_{\mathrm{COP}} P_i^{\mathrm{baseline}}, \tag{23}$$

where P_i^{baseline} is the baseline power of TESS i at time t. Through the above analysis, the baseline power P_i^{baseline} can be expressed as follows:

$$P_i^{\mathrm{baseline}} = \frac{K_i^{\mathrm{h}}\left(T_i^{\mathrm{o}}(t) - T_i^{\mathrm{set}}\right) + \varepsilon_i A_i}{\eta_{\mathrm{COP}}}. \tag{24}$$

The normalized power of TESS i is defined as follows:

$$p_i^{\mathrm{TESS}}(t) = \frac{P_i^{\mathrm{TESS}}(t) - P_i^{\mathrm{baseline}}}{P_i^{\mathrm{TESS,max}}}, \tag{25}$$

where $p_i^{\mathrm{TESS}}(t)$ is the normalized power; $P_i^{\mathrm{TESS,max}}$ is the maximum regulation capacity of TESS i. The SOC of the TESS is defined based on the indoor temperature limit as follows:

$$soc_i^{\mathrm{TESS}}(t) = \frac{T_i^{\max} - T_i^{\mathrm{a}}(t)}{T_i^{\max} - T_i^{\min}}, \tag{26}$$

where $soc_i^{\mathrm{TESS}}(t)$ is the SOC of TESS i at time t; $T_i^{\max}$ and $T_i^{\min}$ are the upper and lower bounds of the indoor temperature, respectively; and $T_i^{\mathrm{a}}(t)$ is the indoor temperature.

4) State-space Model of TESS: Based on the first-order temperature model and the SOC definition, the SOC dynamics of TESS i can be expressed as follows:

$$\dot{soc}_i^{\mathrm{TESS}}(t) = a_i soc_i^{\mathrm{TESS}}(t) + \gamma_i^{\mathrm{TESS}} p_i^{\mathrm{TESS}}(t) + \delta_i^{\mathrm{TESS}}(t). \tag{27}$$

The coefficients γ_i^{TESS} and $\delta_i^{\mathrm{TESS}}(t)$ can be expressed as follows:

$$\begin{cases} \gamma_i^{\mathrm{TESS}} = -\dfrac{c_i P_i^{\mathrm{TESS,max}}}{T_i^{\max} - T_i^{\min}} = \dfrac{\eta_{\mathrm{COP}} P_i^{\mathrm{TESS,max}}}{C_i\left(T_i^{\max} - T_i^{\min}\right)}, \\ \delta_i^{\mathrm{TESS}}(t) = -\dfrac{a_i T_i^{\max} + b_i T_i^{\mathrm{o}}(t) + c_i P_i^{\mathrm{baseline}} + d_i}{T_i^{\max} - T_i^{\min}}. \end{cases} \tag{28}$$

For compact representation, the state of TESS i is defined as follows:

$$\mathbf{x}_i^{\mathrm{TESS}}(t) = \begin{bmatrix} soc_i^{\mathrm{TESS}}(t) \\ p_i^{\mathrm{TESS}}(t) \end{bmatrix}, \tag{29}$$

where $\mathbf{x}_i^{\mathrm{TESS}}(t)$ is the state vector composed of the SOC and normalized power. The state-space and output model of TESS i can be written as

$$\begin{cases} \dot{\mathbf{x}}_i^{\mathrm{TESS}}(t) = \mathbf{A}_i^{\mathrm{TESS}} \mathbf{x}_i^{\mathrm{TESS}}(t) + \mathbf{B}_i^{\mathrm{TESS}} u_i^{\mathrm{TESS}}(t) + \mathbf{d}_i^{\mathrm{TESS}}(t), \\ y_i^{\mathrm{TESS}}(t) = \mathbf{C}_i^{\mathrm{TESS}} \mathbf{x}_i^{\mathrm{TESS}}(t) + E_i^{\mathrm{TESS}}, \end{cases} \tag{30}$$

where $y_i^{\mathrm{TESS}}(t)$ is the output of the TESS i; $u_i^{\mathrm{TESS}}(t)$ is the control input that acts on the rate of change of $p_i^{\mathrm{TESS}}(t)$; $\mathbf{A}_i^{\mathrm{TESS}}$, $\mathbf{B}_i^{\mathrm{TESS}}$, $\mathbf{C}_i^{\mathrm{TESS}}$ and E_i^{TESS} are the coefficient matrices and variables, respectively; $\mathbf{d}_i^{\mathrm{TESS}}(t)$ is the disturbance term that aggregates the effect of outdoor temperature, baseline power and internal heat gain; The $\mathbf{A}_i^{\mathrm{TESS}}$, $\mathbf{B}_i^{\mathrm{TESS}}$, $\mathbf{C}_i^{\mathrm{TESS}}$ and $\mathbf{d}_i^{\mathrm{TESS}}(t)$ can be expressed as follows:

$$\begin{cases} \mathbf{A}_i^{\mathrm{TESS}} = \begin{bmatrix} a_i & \gamma_i^{\mathrm{TESS}} \\ 0 & 0 \end{bmatrix}, \mathbf{B}_i^{\mathrm{TESS}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ \mathbf{C}_i^{\mathrm{TESS}} = \begin{bmatrix} 0 \\ P_i^{\mathrm{TESS,max}} \end{bmatrix}, \mathbf{d}_i^{\mathrm{TESS}}(t) = \begin{bmatrix} \delta_i^{\mathrm{TESS}}(t) \\ 0 \end{bmatrix}, \\ E_i^{\mathrm{TESS}} = P_i^{\mathrm{baseline}}. \end{cases} \tag{31}$$

E. Modelling of PACES

To obtain a unified PACES representation, the BESS and TESS state-space models in Eqs. (12) and (30) are combined. The state of PACES i is defined as follows:

$$\mathbf{x}_i^{\mathrm{PACES}}(t) = \begin{bmatrix} \mathbf{x}_i^{\mathrm{BESS}}(t) \\ \mathbf{x}_i^{\mathrm{TESS}}(t) \end{bmatrix}. \tag{32}$$

The output of PACES i can be expressed as follows:

$$y_i^{\mathrm{PACES}}(t) = y_i^{\mathrm{BESS}}(t) + y_i^{\mathrm{TESS}}(t). \tag{33}$$

The input of PACES i can be expressed as follows:

$$\mathbf{u}_i(t) = \begin{bmatrix} u_i^{\mathrm{BESS}}(t) \\ u_i^{\mathrm{TESS}}(t) \end{bmatrix}. \tag{34}$$

The disturbance of PACES i can be expressed as follows:

$$\mathbf{d}_i(t) = \begin{bmatrix} \mathbf{d}_i^{\mathrm{BESS}} \\ \mathbf{d}_i^{\mathrm{TESS}}(t) \end{bmatrix} = \begin{bmatrix} \delta_i^{\mathrm{BESS}} \\ 0 \\ \delta_i^{\mathrm{TESS}}(t) \\ 0 \end{bmatrix}. \tag{35}$$

The aggregated state-space and output model of all PACES can be expressed as follows:

$$\begin{cases} \dot{\mathbf{X}}(t) = \mathbf{A}\mathbf{X}(t) + \begin{bmatrix} \mathbf{B}^{\mathrm{BESS}} \\ \mathbf{B}^{\mathrm{TESS}} \end{bmatrix} \left[\, \mathbf{U}^{\mathrm{BESS}}(t)\ \mathbf{U}^{\mathrm{TESS}}(t) \,\right] + \mathbf{D}(t), \\ \mathbf{Y}(t) = \mathbf{C}\mathbf{X}(t) + \mathbf{E}(t), \end{cases} \tag{36}$$

where $\mathbf{X}(t)$ is the global state of the state equation; $\mathbf{Y}(t)$ is the global output of the PACES; $\mathbf{U}^{\mathrm{BESS}}(t)$ is the BESS input vector; $\mathbf{U}^{\mathrm{TESS}}(t)$ is the TESS input vector; $\mathbf{D}(t)$ is the disturbance matrix; $\mathbf{A}$, $\mathbf{B}^{\mathrm{BESS}}$, $\mathbf{B}^{\mathrm{TESS}}$ and $\mathbf{C}(t)$ are the

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coefficient matrices of the state equation. The above matrices and vectors can be specifically denoted as follows:

$$\begin{cases} \mathbf{X}(t) = \left[\begin{array}{cccc} \mathbf{x}_1^{\mathrm{T}}(t) & \mathbf{x}_2^{\mathrm{T}}(t) & \cdots & \mathbf{x}_I^{\mathrm{T}}(t) \end{array}\right]^{\mathrm{T}}, \\ \mathbf{Y}(t) = \left[\begin{array}{cccc} y_1(t) & y_2(t) & \cdots & y_I(t) \end{array}\right]^{\mathrm{T}}, \\ \mathbf{U}^{\mathrm{BESS}}(t) = \left[\begin{array}{cccc} u_1^{\mathrm{BESS}}(t) & \cdots & u_I^{\mathrm{BESS}}(t) \end{array}\right]^{\mathrm{T}}, \\ \mathbf{U}^{\mathrm{TESS}}(t) = \left[\begin{array}{cccc} u_1^{\mathrm{TESS}}(t) & \cdots & u_I^{\mathrm{TESS}}(t) \end{array}\right]^{\mathrm{T}}, \\ \mathbf{A} = \mathrm{diag}(\mathbf{A}_1, \ldots, \mathbf{A}_I), \\ \mathbf{A}_i = \begin{bmatrix} \mathbf{A}_i^{\mathrm{BESS}} & 0 \\ 0 & \mathbf{A}_i^{\mathrm{TESS}} \end{bmatrix}, \\ \mathbf{B}^{\mathrm{BESS}} = \mathrm{diag}(\mathbf{B}_1^{\mathrm{BESS}}, \ldots, \mathbf{B}_I^{\mathrm{BESS}}), \\ \mathbf{B}^{\mathrm{TESS}} = \mathrm{diag}(\mathbf{B}_1^{\mathrm{TESS}}, \ldots, \mathbf{B}_I^{\mathrm{TESS}}), \\ \mathbf{C} = \mathrm{diag}(\mathbf{C}_1, \ldots, \mathbf{C}_I), \\ \mathbf{C}_i = \left[\begin{array}{cc} \mathbf{C}_i^{\mathrm{BESS}} & \mathbf{C}_i^{\mathrm{TESS}} \end{array}\right], \\ \mathbf{D}(t) = \left[\begin{array}{cccc} \mathbf{d}_1^{\mathrm{T}}(t) & \mathbf{d}_2^{\mathrm{T}}(t) & \cdots & \mathbf{d}_I^{\mathrm{T}}(t) \end{array}\right]^{\mathrm{T}}. \\ \mathbf{E}(t) = \mathrm{diag}(E_1^{\mathrm{TESS}}, \ldots, E_I^{\mathrm{TESS}}). \end{cases} \tag{37}$$

III. Distributed Consensus Control of Multiple PACES

A. Power Regulation Objective

Multiple PACES may exhibit different dynamic responses during frequency regulation because their BESS, TESS, and building thermal parameters are heterogeneous. Therefore, a distributed consensus strategy is adopted to coordinate these heterogeneous PACES through local information exchange among neighboring nodes. As shown in Fig. 2, the control objective is to make the aggregated PACES power $P_{\mathrm{REG}}(t)$ track the regulation demand $P_{\mathrm{DEM}}(t)$, thereby reducing the remaining power imbalance. To quantify the deviation between $P_{\mathrm{DEM}}(t)$ and $P_{\mathrm{REG}}(t)$, the global regulation error is defined as follows:

$$e(t) = P_{\mathrm{DEM}}(t) - P_{\mathrm{REG}}(t), \tag{38}$$

where $e(t)$ represents the remaining regulation error that needs to be compensated through the coordinated BESS-TESS response of PACES.

B. Distributed Consensus-based Controller

The information exchange among multiple PACES is represented by a connected communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node denotes one PACES and each edge denotes neighbor-to-neighbor communication. The communication graph is described by the Laplacian matrix $\mathbf{L} \in \mathbb{R}^{I \times I}$ [41]. The element a_{ij} denotes the communication weight between PACES i and PACES j, and the neighbor set of PACES i is denoted by $\mathcal{N}_i$. Based on the neighbor information and the global regulation error $e(t)$, the distributed consensus controller updates the local regulation signal of each PACES. For each PACES i, the local regulation signal is updated as follows:

$$\dot{y}_i(t) = -k_\theta \sum_{j \in \mathcal{N}_i} a_{ij}\big(y_i(t) - y_j(t)\big) + k_e b_i\, e(t), \tag{39}$$

where k_θ is the consensus coupling gain that drives neighboring PACES to reach agreement, $k_\theta > 0$; k_e is the regulation gain that determines how the global regulation error is injected into the DSO-connected PACES, $k_e > 0$; b_i denotes whether PACES i directly receives the DSO signal, $b_i \in \{0, 1\}$. In the proposed controller, $b_i = 1$ for the DSO-connected PACES, while $b_i = 0$ for the remaining PACES. Therefore, only the DSO-connected PACES directly receive the global regulation error, and the remaining PACES update their local regulation signals through neighbor-to-neighbor information exchange.

Algorithm 1: Workflow of the Proposed PACES Distributed Consensus Control.

Input: P_{GRID}, P_{LOAD}, P_i^{PV}, P_i^{BESS}, P_i^{TESS}, SOC_i^{BESS}, soc_i^{TESS}, T_i^{a}, $\mathcal{N}_i$, $\mathbf{L}$, $\mathbf{b}$.
Output: $P_i^{\mathrm{BESS,ref}}$, $P_i^{\mathrm{TESS,ref}}$, P_{REG}.

```
for t = 1, 2, ... do
    P_DEM ← P_GRID − P_LOAD − Σ_{i=1}^{I} P_i^PV;
    P_REG ← Σ_{i=1}^{I} (P_i^BESS + P_i^TESS);
    e ← P_DEM − P_REG;
    for i = 1, 2, ..., I do
        Receive y_j from j ∈ N_i;
        ẏ_i ← −k_θ Σ_{j∈N_i} a_ij (y_i − y_j) + k_e b_i e;
        Estimate available TESS regulation under
          T_i^min ≤ T_i^a ≤ T_i^max and TESS SOC/power
          limits;
        P_i^TESS,ref ← available TESS part of y_i;
        P_i^BESS,ref ← y_i − P_i^TESS,ref under BESS
          SOC/power limits;
    Update P_i^BESS, P_i^TESS, and P_REG;
```

By stacking the local update equations of all PACES in Eq. (39), the compact global representation can be obtained as follows:

$$\dot{\mathbf{Y}}(t) = -k_\theta \mathbf{L}\mathbf{Y}(t) + k_e \mathbf{b}\, e(t), \tag{40}$$

where $\mathbf{b} = [b_1, \ldots, b_I]^\top$. The consensus-updated signal $y_i(t)$ serves as the node-level regulation reference for PACES i. After receiving this reference, each PACES coordinates its internal BESS and TESS response while satisfying the SOC and indoor-temperature constraints. In this way, the proposed controller converts the global regulation error into local PACES regulation references through neighbor-to-neighbor communication, which enables coordinated frequency regulation without requiring every PACES to directly receive the DSO signal [42]. To make the control workflow clearer, Algorithm 1 summarizes how the system-level regulation demand is calculated, how the global regulation error is formed, and how the node-level PACES regulation signals are updated through distributed consensus.

IV. Experiment and Result Analysis

A. Experiment Setup

Fig. 3 shows the RTDS-based hardware-in-the-loop experiment platform. RTDS simulates the power system connected with PACES, while the PACES model and the proposed distributed consensus controller are implemented on the PC

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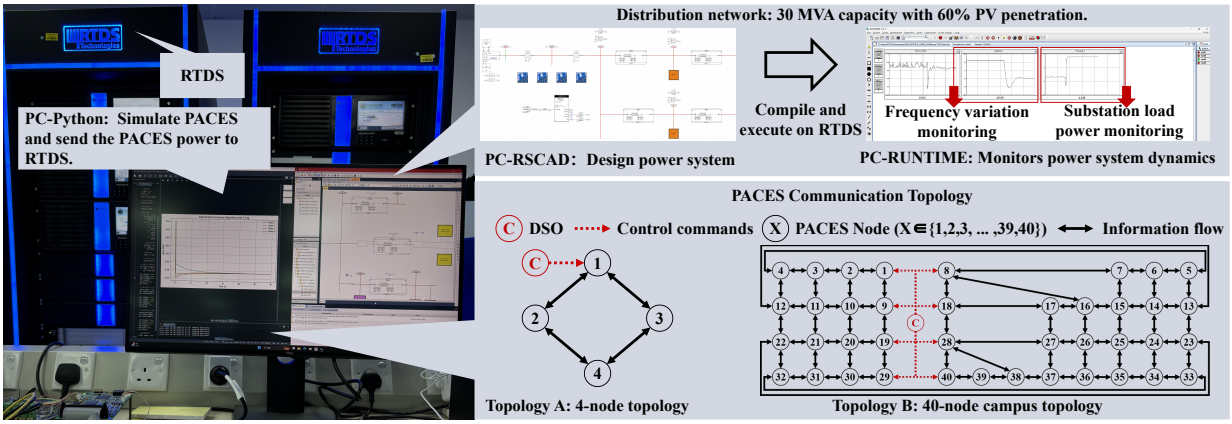


Fig. 3. Hardware-in-the-loop experiment platform.

side through Python. RTDS simulates a DN station area with a capacity of 30 MVA, where 60% of the power is generated from PV, including centralized and distributed PV. During the experiment, the PC receives real-time DN frequency and load power information from RTDS, together with PACES-side operating information including PV power, BESS power and SOC, TESS power, indoor/outdoor temperature, and the temperature setpoint of the air conditioner, calculates the PACES power response based on Eqs. (4)–(37), and sends the BESS/TESS power references back to RTDS. The neighbor-to-neighbor information exchange among PACES represents the communication condition required by the proposed distributed consensus controller, while the indoor-temperature constraints in the TESS model are used to maintain occupant comfort. This closed-loop process evaluates the interaction between PACES and the DN under real-time power system dynamics. Table I lists the parameters of PACES.

To validate the proposed consensus control algorithm, a 4-node typical PACES topology is first employed, as shown in Fig. 3. Each PACES has a PV peak power of 1.0 MW, a BESS capacity of 1.5 MWh, and a TESS rated power of 0.6 MW. In this topology, PACES 1 is selected as the leader node that directly receives the regulation signal from the DSO. Accordingly, b_1 is set to 1, while b_i is set to 0 for the other PACES. PACES 1 exchanges information with PACES 2 and PACES 3, and PACES 4 receives the regulation information through PACES 2 and PACES 3. The control step is set to 0.1 s, and the PV-induced disturbance is introduced at 5 s in the plotted time window. A real campus-derived 40-node topology is further used to evaluate the scalability and communication robustness of the proposed method, as shown in Fig. 3. In this topology, only the DSO-connected PACES directly receive the regulation signal, while the remaining PACES update their local regulation references through neighbor-to-neighbor communication. The average computation time of the proposed consensus update is about 0.02 ms per control step, which is much shorter than the 0.1 s control step. The detailed settings and results are presented in the following subsections.

B. Result Analysis of Frequency Regulation under the 4-node Typical Topology

To verify the effectiveness of the proposed regulation method and compare it with the centralized droop benchmark,

TABLE I
THE PARAMETERS FOR THE PACES

Symbols	Values	Units	Symbols	Values	Units
c_i^{a}	1.005	kJ/(kg·°C)	$\tau_{c,i}$	0.02	s
ρ_i^{a}	1.205	kg/m^3	α_i	2.0	Hz/°C
A_i	$\mathcal{U}(100,120)\times 100$	m^2	β_i	0.005	Hz/(°C·s)
A_i^{S}	$\mathcal{U}(40,60)\times 100$	m^2	$f_i^{\mathrm{TESS,min}}$	$\mathcal{U}(5,15)$	Hz
V_i	$\mathcal{U}(300,360)\times 100$	m^3	$f_i^{\mathrm{TESS,max}}$	$\mathcal{U}(135,150)$	Hz
ζ_i	0.5	1/hour	$\mathrm{SOC}_i^{\min}$	$\mathcal{U}(0.05,0.2)$	–
U_i^{h}	$\mathcal{U}(0.1,8.0)$	W/(m^2 ·° C)	$\mathrm{SOC}_i^{\max}$	$\mathcal{U}(0.8,0.95)$	–
T_i^{o}	32	°C	P_i^{d}	$\mathcal{U}(400,600)$	kW
T_i^{a}	24 ($t=0$)	°C	P_i^{c}	$\mathcal{U}(400,600)$	kW
κ_i^{p}	0.04	kW/Hz	μ_i^{p}	0.02	kW

$\mathcal{U}$ denotes uniform distribution.
A_i and A_i^{S} denote the equivalent floor area and the equivalent envelope area for outdoor heat exchange, respectively.

TABLE II
CONTROL STRATEGIES IN DIFFERENT CASES.

Case	Controller	Regulation Resource
Case 0	No regulation	Baseline without PACES regulation
Case 1	Centralized droop benchmark	BESS-only regulation
Case 2	Centralized droop benchmark	BESS-TESS coordination
Case 3	Proposed distributed consensus	BESS-only regulation
Case 4	Proposed distributed consensus	BESS-TESS coordination
Case 5	Proposed distributed consensus	TESS rebound compensation by BESS

the case-comparison experiment is designed as shown in Case 0 to Case 4 in Table II. Case 0 is the baseline without PACES regulation. Case 1 and Case 2 are centralized droop benchmark cases. In these two cases, the regulation command is available to the central controller and directly allocated to all PACES according to the PACES regulation capability. Since the regulation capacities of PACES are identical in this 4-node topology, the centralized droop allocation becomes an equal-sharing benchmark. Case 1 uses BESS-only regulation, while Case 2 uses BESS-TESS coordination. Case 3 and Case 4 use the proposed distributed consensus control. In these two cases, the regulation command is injected only into the leader PACES, and the remaining PACES update their power references through neighbor-to-neighbor communication. Case 3 uses BESS-only regulation, while Case 4 uses BESS-TESS coordination. These cases serve as reference cases to verify the advantages of the proposed method in frequency support, distributed coordination, and BESS-TESS regulation sharing. The experiment is designed for a 1.8 MW power deficit in the DN after 5 s of system operation, which is caused by PV fluctuations.

Fig. 4 shows the DN frequency response, aggregated PACES

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TABLE III
MAIN EVALUATION INDICATORS IN THE 4-NODE CASE-COMPARISON EXPERIMENT.

Case	Freq. Nadir (Hz)	Final Power (MW)	Tracking RMSE (MW)	Settling Time (s)	SOC Range	Reg. Throughput (MWh)
Case 0	49.769	–	–	–	–	–
Case 1	49.925	0.2	0.292	4.4	0.2337	0.02585
Case 2	49.925	0.2	0.292	4.4	0.1816	0.00804
Case 3	49.925	0.2	0.292	4.4	0.2334	0.02585
Case 4	49.925	0.2	0.292	4.4	0.1813	0.00804

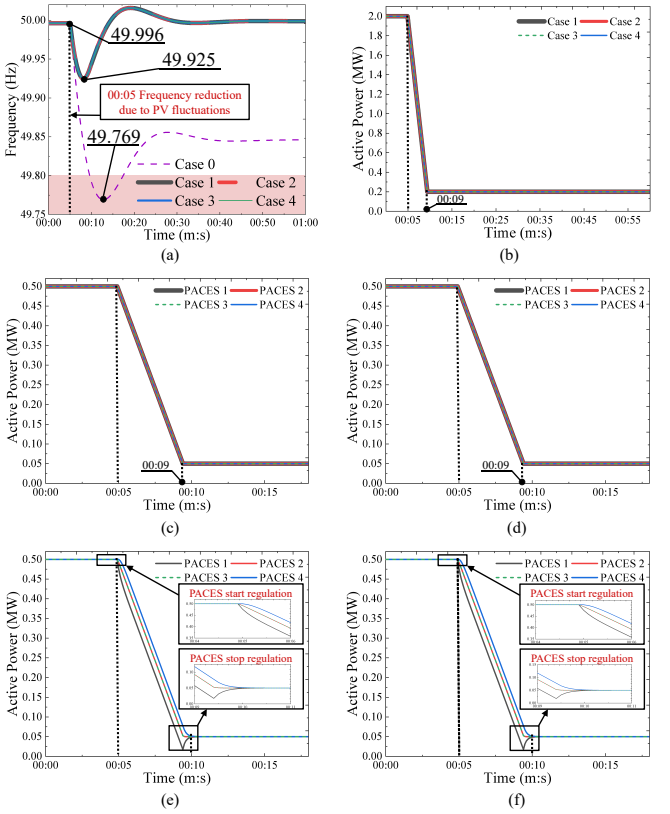


Fig. 4. Frequency regulation and PACES power responses under a 1.8 MW power deficit caused by PV power fluctuations: (a) Frequency deviation of the distribution network; (b) Aggregated power variation of all PACESs; (c) Each PACES power variation in Case 1; (d) Each PACES power variation in Case 2; (e) Each PACES power variation in Case 3; (f) Each PACES power variation in Case 4.

power response, and individual PACES power responses under Case 0 to Case 4. The main evaluation indicators are summarized in Table III. The frequency-regulation and power-tracking performance are mainly evaluated using frequency nadir, final power, tracking RMSE, and settling time. The frequency nadir is the lowest frequency reached after the PV-induced disturbance and reflects the most severe frequency drop. The final power is the final aggregated PACES power after regulation. The tracking RMSE evaluates the deviation between the aggregated PACES power and the target power during the regulation process. The settling time is the time required for the aggregated PACES power to remain within the 2% band of the target value.

As shown in Fig. 4(a), the pre-disturbance DN frequency is 49.996 Hz. Without PACES regulation in Case 0, the frequency nadir decreases to 49.769 Hz. In contrast, with PACES regulation in Case 1 to Case 4, the frequency nadir is improved to 49.925 Hz. Therefore, the frequency deviation relative to the nominal 50 Hz is reduced from 0.231 Hz to 0.075 Hz, and the disturbance-induced additional frequency drop relative to 49.996 Hz is reduced from 0.227 Hz to 0.071 Hz. The regulated frequency deviation is also within the ± 0.2 Hz reference range specified in GB/T 15945-2008 [43] for 50 Hz power systems. These results show that PACES can provide effective frequency support under the 1.8 MW PV-induced power deficit. As shown in Fig. 4(b), Case 1 to Case 4 all reduce the aggregated PACES power from 2.0 MW to the target value of 0.2 MW, which compensates for the 1.8 MW power deficit. Fig. 4(c)–(f) further show the individual PACES power responses. In the centralized droop benchmark cases, the regulation command is directly allocated to all PACES. In the proposed distributed consensus cases, PACES 1 first receives the regulation signal, PACES 2 and PACES 3 update their power references through communication with PACES 1, and PACES 4 receives the regulation information through PACES 2 and PACES 3. The results demonstrate that the proposed distributed consensus control can achieve aggregated power tracking through local neighbor-to-neighbor communication. As shown in Table III, Case 1 to Case 4 have the same frequency nadir, final aggregated power, tracking RMSE, and settling time. This indicates that the proposed distributed consensus control achieves comparable frequency-support and aggregated power-tracking performance to the centralized droop benchmark.

C. Result Analysis of PACES During the Frequency Regulation

To analyze the heterogeneous regulation capabilities of BESS and TESS, Fig. 5(a)–(d) compares the dynamic responses in Case 3 and Case 4, respectively. Case 3 uses BESS-only regulation, while Case 4 uses BESS-TESS coordination under the proposed distributed consensus control. As shown in Fig. 5(a), BESS provides fast electrical power support in both cases. In Case 3, the regulation demand is mainly supplied by BESS, which leads to a continuous decrease in BESS SOC, as shown in Fig. 5(b). In Case 4, TESS participates in regulation by adjusting the air conditioner operating state, and BESS compensates the remaining regulation demand. Therefore, the BESS SOC changes within a smaller range in Case 4. Meanwhile, Fig. 5(d) shows that the indoor temperature deviation remains about 1 °C, which indicates that TESS participation is achieved within the thermal comfort constraint.

The last two columns of Table III further quantify the influence of TESS participation on BESS operation. The SOC range is calculated as the maximum BESS SOC minus the minimum BESS SOC during the regulation process. A smaller SOC range means that the battery experiences a

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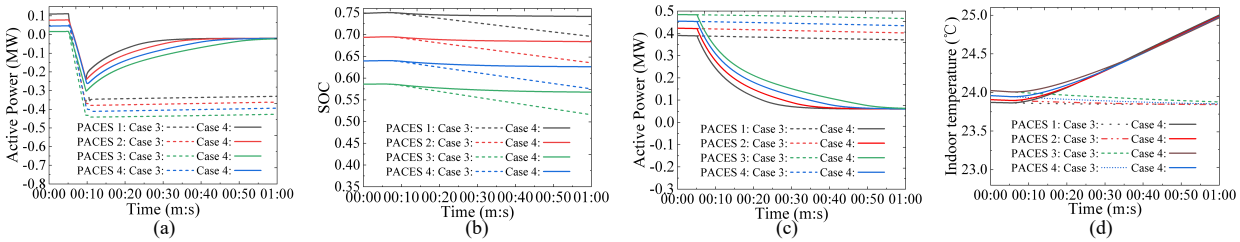


Fig. 5. Dynamic response of BESS and TESS in Case 3 and Case 4, respectively: (a) BESS power variation; (b) BESS SOC variation; (c) TESS power variation; (d) Temperature variation.

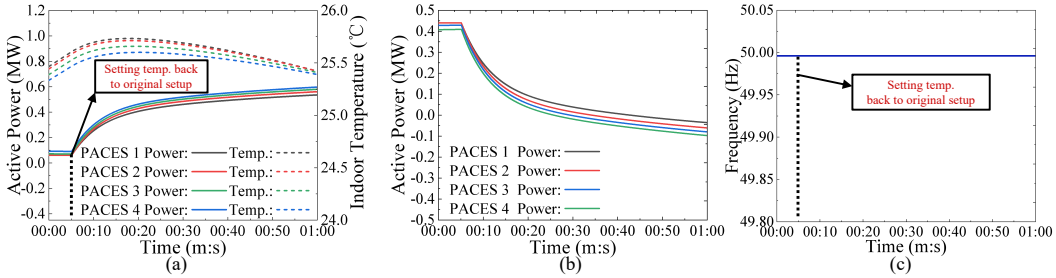


Fig. 6. Experimental results of TESS power rebound and BESS compensation in PACES at the end of frequency regulation: (a) TESS power and temp. variation in each PACES; (b) BESS power variation in each PACES; (c) Frequency response of the DN.

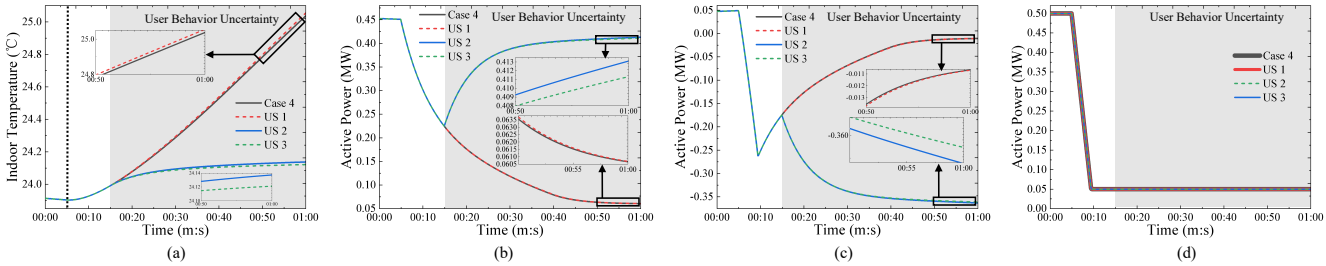


Fig. 7. Dynamic responses of PACES 2 under user behavior uncertainty: (a) PACES 2 indoor temperature; (b) PACES 2 TESS power; (c) PACES 2 BESS power; (d) PACES 2 total power.

smaller charge/discharge depth. The BESS regulation energy throughput is calculated as the time integral of the absolute BESS power variation relative to the pre-disturbance operating point. A smaller regulation energy throughput means that BESS delivers or absorbs less additional regulation energy during the same frequency-support process. Compared with Case 3, Case 4 reduces the BESS SOC range from 0.2334 to 0.1813 and reduces the BESS regulation energy throughput from 0.02585 MWh to 0.00804 MWh. These results indicate that TESS participation transfers part of the regulation burden from BESS to air conditioner flexibility, which helps reduce BESS cycling stress while maintaining the frequency-regulation performance.

D. Result Analysis of PACES at the End of Frequency Regulation

To analyze the influence of air conditioner temperature-control restoration at the end of frequency regulation, Case 5 in Table II is considered. In this case, the TESS setting temperature returns to the original setup at 5 s after the frequency regulation process, while PACES remains under the proposed distributed consensus control.

As shown in Fig. 6(a), when the TESS setting temperature returns to the original setup, the TESS operating power gradually increases as the indoor temperature is restored toward the original comfort condition. This TESS power rebound introduces an additional load variation to the DN. As shown in Fig. 6(b), the proposed consensus control coordinates BESS to provide compensating power support for the TESS power rebound. Therefore, the DN frequency remains stable without an additional frequency drop, as shown in Fig. 6(c). This result demonstrates that the proposed BESS-TESS coordination can compensate for the rebound power at the end of frequency regulation.

E. User Behavior Uncertainty Experiment Under the 4-node Typical Topology

To further examine the robustness of the proposed BESS-TESS coordination under user behavior uncertainty, an additional 4-node experiment is conducted based on Case 4 in Table II. Three user behavior uncertainty scenarios are considered for PACES 2, as defined in Table IV. The uncertainty events start 10 s after the regulation command is activated and remain until the end of the observation window. The other PACES settings and communication topology remain the same as those in Case 4.

Fig. 7 shows the dynamic responses of PACES 2 under Case 4 and the three user behavior uncertainty scenarios defined in Table IV. As shown in Fig. 7(a), the internal heat-gain increase in US1 slightly increases the indoor temperature compared with Case 4. In US2 and US3, the user-side setpoint restoration changes the indoor thermal trajectory more significantly because PACES 2 no longer fully follows the demand-response setting. As shown in Fig. 7(b), the TESS power

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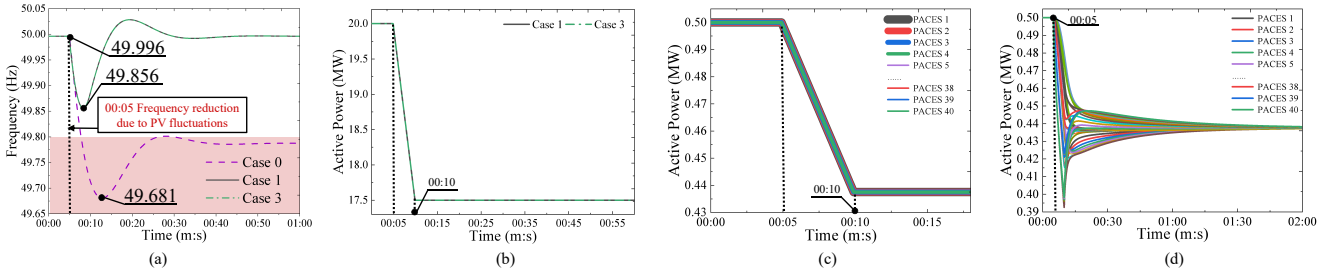


Fig. 8. Frequency and PACES power responses under the 40-node campus topology: (a) Frequency deviation of the distribution network; (b) Aggregated power variation of all PACESs; (c) Each PACES power variation in Case 1; (d) Each PACES power variation in Case 3.

TABLE IV
MAIN EVALUATION INDICATORS UNDER USER BEHAVIOR UNCERTAINTY IN THE 4-NODE EXPERIMENT.

Scenario	Setting	Freq. Nadir (Hz)	Final Agg. Power (MW)	Settling Time (s)
Case 4	Deterministic case	49.925	0.2	4.4
US1	Heat-gain increase	49.925	0.2	4.4
US2	Setpoint change	49.925	0.2	4.4
US3	Combined uncertainty	49.925	0.2	4.4

increases under US2 and US3 because the air conditioner restores the indoor temperature toward the original comfort setting. Correspondingly, Fig. 7(c) shows that BESS adjusts its power in the opposite direction to compensate for the TESS power variation. As a result, the total power of PACES 2 remains close to the Case 4 response as shown in Fig. 7(d). The main global regulation indicators are summarized in Table IV. The four scenarios have the same frequency nadir, final aggregated power, and settling time. These results indicate that user behavior uncertainty changes the internal TESS-BESS sharing of PACES 2, while the proposed coordination maintains the required PACES regulation response.

F. Scalability Validation Under the 40-node Campus Topology

To evaluate scalability, a campus-derived 40-node topology is used for extended validation, as shown in Fig. 3. This setting verifies whether the proposed strategy can propagate the regulation command and maintain aggregated power tracking in a larger PACES group. In the 40-node experiment, the initial aggregated PACES power is 20 MW. After the PV-induced power imbalance occurs at 5 s, the target aggregated PACES power is set to 17.5 MW, corresponding to a total regulation requirement of 2.5 MW.

As shown in Fig. 8(a), without PACES regulation, the DN frequency decreases significantly after the disturbance, and the frequency nadir is about 49.681 Hz. With PACES regulation, the frequency nadir is improved to about 49.856 Hz. As shown in Fig. 8(b), the aggregated PACES power is reduced from 20 MW to the target value of 17.5 MW, which provides the required 2.5 MW power reduction. Fig. 8(c) and Fig. 8(d) further show the individual PACES power responses under the centralized benchmark and the proposed distributed control, respectively. These results indicate that the proposed distributed consensus control can extend from the 4-node typical topology to the 40-node campus topology while maintaining aggregated power tracking and frequency-support capability. It should be noted that the communication burden of the proposed distributed controller is mainly determined by the local communication degree rather than the total number of PACES. In the adopted neighbor-to-neighbor communication architecture, each PACES exchanges information only with its adjacent PACES. For example, in the tested topologies, each PACES communicates with no more than four neighboring PACES. Therefore, as the PACES scale increases, the communication requirement of each terminal remains bounded by the local communication degree, without requiring each PACES to communicate with all other PACES.

G. Long-Horizon Validation Under Time-Varying Regulation Targets

To validate longer-horizon operation under time-varying PV-induced power imbalance conditions, a 4-hour RTDS-based HIL experiment is conducted under the 40-node campus topology. In this experiment, the regulation target is updated every 15 min and varies within 18.5–21.5 MW around the 20 MW baseline. Both upward and downward regulation target changes are included to emulate sequential PV-induced power surplus and deficit conditions. A downward target change corresponds to PACES power reduction for compensating a PV-induced power deficit, whereas an upward target change corresponds to PACES power increase for absorbing PV power surplus.

As shown in Fig. 9(a), the aggregated PACES power closely tracks the time-varying regulation target during the 4-hour RTDS-based HIL experiment. Fig. 9(b) shows the corresponding RTDS frequency response. The frequency deviations are caused by the time-varying power imbalance induced by PV power fluctuations. The DSO converts this imbalance into upward or downward PACES regulation targets, and the coordinated PACES response is used to compensate for the corresponding power surplus or deficit. When the regulation target decreases, PACES reduces its aggregated power to compensate for a power deficit and support the frequency recovery. When the regulation target increases, PACES increases its aggregated power to absorb surplus power and suppress the frequency rise. Therefore, the frequency deviations are transient, and the frequency returns close to 50 Hz after PACES regulation. The frequency remains within 49.8–50.2 Hz, and Fig. 9(c) further shows that the indoor temperatures of the 40 PACES are maintained within ± 1 °C around the setpoint during the long-horizon experiment. These results demonstrate that the proposed distributed BESS-TESS coordination can maintain aggregated power tracking, frequency support, and indoor-temperature constraints under time-varying regulation targets over a longer operating horizon.

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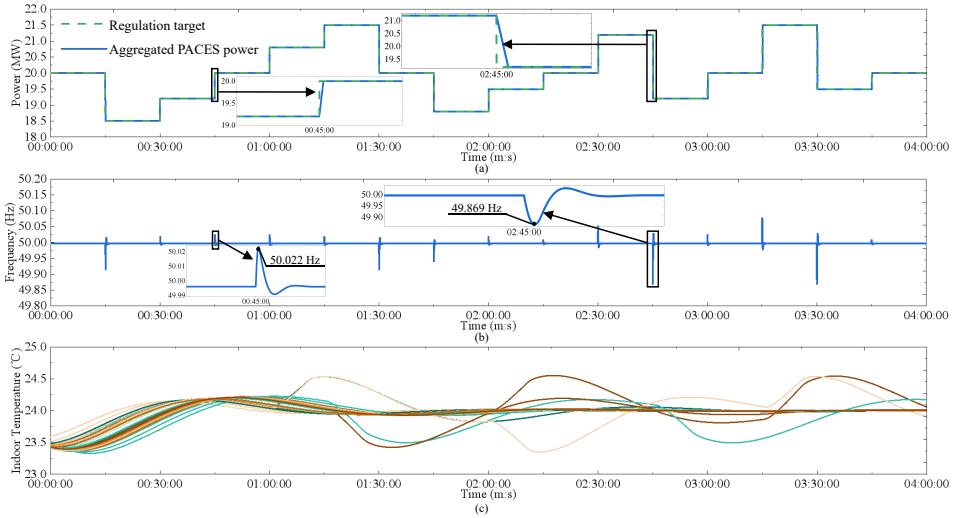


Fig. 9. 4-hour RTDS-based HIL validation under time-varying upward and downward regulation targets: (a) Aggregated PACES power tracking; (b) Frequency response; (c) Indoor temperature responses of 40 PACES.

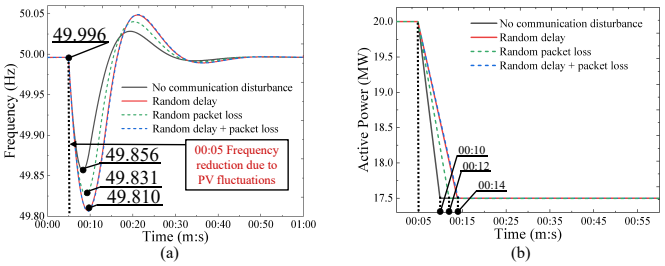


Fig. 10. Communication-robustness validation under the 40-node campus topology: (a) Frequency deviation of the distribution network; (b) Aggregated power variation of all PACESs.

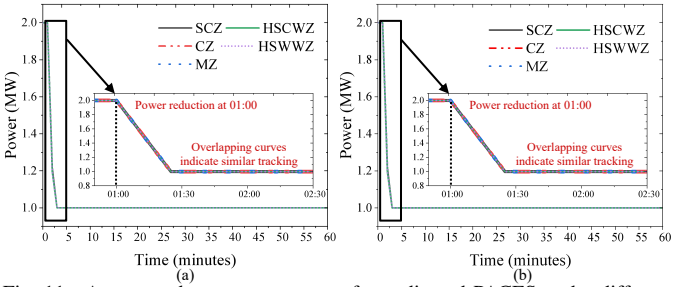


Fig. 11. Aggregated power responses of coordinated PACES under different building climate zones: (a) Aggregated power of PACESs in Case 3; (b) Aggregated power of PACESs in Case 4.

H. Communication Robustness Under Random Delay and Packet Loss

To evaluate the influence of non-ideal IoT communication, communication-robustness experiments are conducted under the 40-node campus topology. The initial aggregated PACES power is 20 MW, and the target value is 17.5 MW after the PV-induced disturbance, corresponding to a 2.5 MW power reduction. Four communication scenarios are considered: no communication disturbance, random delay, random packet loss, and random delay with packet loss. In the random delay scenario, the neighbor-to-neighbor information update is delayed by 0–2 control steps, corresponding to 0–0.2 s. In the random packet loss scenario, the packet loss rate is randomly generated within 0–10%, and the latest successfully received value is used when packet loss occurs.

As shown in Fig. 10(a), communication disturbances affect the transient frequency response. Without communication disturbance, the frequency nadir is about 49.856 Hz, which is consistent with the 40-node validation result in Fig. 8(a). Under random packet loss, the frequency nadir decreases to about 49.831 Hz. Under random delay and the combined delay-loss disturbance, the frequency nadir further decreases to about 49.810 Hz. This result indicates that random delay has a stronger influence on the transient frequency response than packet loss because delayed neighbor information slows down the consensus update. As shown in Fig. 10(b), the aggregated PACES power still converges to the target value of 17.5 MW under all four communication scenarios. The target value is reached at about 10 s without communication disturbance, about 12 s under random packet loss, and about 14 s under random delay or the combined delay-loss disturbance. These results indicate that communication delay and packet loss affect the frequency nadir and response timing, while the proposed distributed consensus control maintains acceptable regulation capability.

TABLE V
CLIMATE-RELATED PARAMETERS FOR THE FIVE CHINESE BUILDING CLIMATE ZONES.

Climate zone	City	T_i^{o} (°C)	U_i^{h} (W/(m^2·°C))	ζ_i (1/hour)
SCZ	Harbin	$\mathcal{U}$(26, 32)	$\mathcal{U}$(0.35, 0.70)	$\mathcal{U}$(0.20, 0.50)
CZ	Beijing	$\mathcal{U}$(28, 35)	$\mathcal{U}$(0.50, 1.00)	$\mathcal{U}$(0.30, 0.70)
MZ	Kunming	$\mathcal{U}$(25, 30)	$\mathcal{U}$(0.60, 1.20)	$\mathcal{U}$(0.40, 0.90)
HSCWZ	Shanghai	$\mathcal{U}$(30, 37)	$\mathcal{U}$(1.20, 2.20)	$\mathcal{U}$(0.60, 1.20)
HSWWZ	Guangzhou	$\mathcal{U}$(31, 38)	$\mathcal{U}$(1.50, 2.80)	$\mathcal{U}$(0.80, 1.50)

$\mathcal{U}$ denotes uniform distributions.
U_i^{h} is related to R_i, and ζ_i is related to C_i.

I. Sensitivity Analysis of Coordinated PACES Regulation under Different Building Climate Zones

The proposed consensus-based coordination method for PACES is evaluated under the five typical Chinese building climate zones in summer. According to China's code for thermal design of civil buildings (GB 50176-2016) [33] and the related building climate classification study [34], Harbin, Beijing, Kunming, Shanghai, and Guangzhou are selected as representative cities for the SCZ, CZ, MZ, HSCWZ, and HSWWZ,

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respectively. This sensitivity analysis is conducted under the 4-node typical topology, and the reported BESS/TESS regulation shares are calculated from the aggregated regulation energy of the four PACES. The key climate-related parameters adopted for the simulations are summarized in Table V, including the outdoor temperature T_i^{o}, the overall heat transfer coefficient between the indoor and outdoor air U_i^{h}, and the air change rate ζ_i. The parameter ranges are used to represent typical summer conditions and bounded building heterogeneity within each climate zone. For each building climate zone, these parameters are used to update the room thermal model and the baseline TESS load, while all the other PACES parameters remain the same as those listed in Table I.

The experiment is designed for a 1 MW power deficit in the DN after 1 minute of system operation, reducing the aggregated PACES power command from 2 MW to 1 MW. Fig. 11 shows that the aggregated power of coordinated PACES under all climate zones closely overlaps with the control command in both Case 3 and Case 4. The overlapping curves indicate that the proposed coordination method can achieve similar aggregated power tracking under different building climate zones. This result demonstrates that the consensus-based controller is robust to variations in outdoor temperature T_i^{o}, heat-transfer coefficient of the building envelope U_i^{h}, and air change rate ζ_i.

Fig. 12 presents the aggregated quantitative regulation effort of BESS and TESS under different building climate zones. Case 3 is mainly supported by BESS. As shown in Fig. 12(a), the TESS share ranges from -49.5% to 21.3%, while the BESS share ranges from 78.7% to 149.5%. In SCZ, the negative TESS share indicates that TESS absorbs energy over the regulation horizon because of heating demand and strong building heat loss. In Case 4, the regulation sharing pattern changes significantly, as shown in Fig. 12(b). SCZ remains BESS-dominated, while CZ, MZ, HSCWZ, and HSWWZ become TESS-dominated. The TESS share reaches 85.6% in CZ, 94.6% in MZ, 111.7% in HSCWZ, and 114.0% in HSWWZ. The negative BESS energy in HSCWZ and HSWWZ indicates that BESS is charged during regulation because the TESS power reduction is larger than the required compensation.

In summary, the building climate zone affects the regulation sharing between BESS and TESS. In hot-summer zones such as HSCWZ and HSWWZ, BESS should retain sufficient charging capability and SOC upper headroom to absorb surplus power induced by large TESS-based load reduction. In contrast, in SCZ and partly CZ, where TESS contribution is limited, BESS requires sufficient energy capacity and discharge margin to sustain regulation. Therefore, building climate zone diversity should be considered when coordinating BESS and TESS in PACES for DN frequency regulation. Although the sensitivity analysis uses Chinese building climate zones as representative cases, the proposed framework is parameterized by climate-related building variables and can be extended to other regions by updating the outdoor temperature, building envelope heat-transfer coefficient, air change rate, and local indoor-temperature constraints.

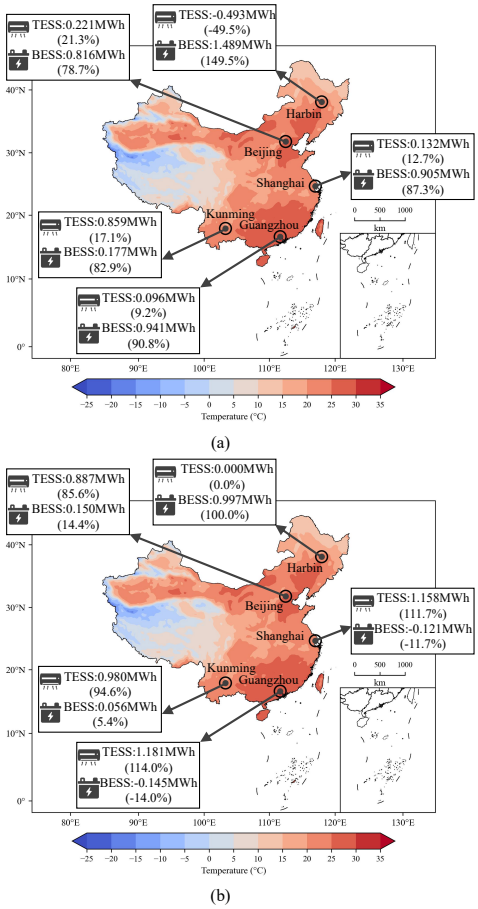


Fig. 12. Quantitative indices of PACES regulation under different building climate zones [35]: (a) Regulation effort of PACES under different building climate zones of Case 3; (b) Regulation effort of PACES under different building climate zones of Case 4.

V. Conclusion

This research provides a climate-aware PACES control framework for coordinating heterogeneous BESS and TESS resources in power system frequency regulation. The innovation of this study is to embed building climate zone diversity into the unified BESS-TESS modeling and distributed PACES coordination. The experiments evaluate frequency regulation performance, BESS-TESS coordination, user behavior uncertainty response, TESS rebound, scalability, communication robustness, and climate-zone adaptability. The main conclusions are summarized as follows:

- The RTDS-based hardware-in-the-loop results demonstrate that PACES can provide effective frequency support under PV-induced power fluctuations. In the tested 4-node case-comparison experiment, the frequency nadir is improved from 49.769 Hz without PACES regulation to 49.925 Hz with PACES regulation, and the corresponding frequency deviation relative to the nominal 50 Hz is reduced from 0.231 Hz to 0.075 Hz. The proposed distributed consensus control achieves comparable aggregated power-tracking performance to the centralized droop benchmark while reducing dependence on full centralized command allocation.
- BESS-TESS coordination reduces the regulation burden of BESS while maintaining comparable frequency-regulation performance under the tested conditions. The

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user behavior uncertainty experiment shows that local changes in internal heat gain and user-side setpoint restoration mainly affect the internal BESS-TESS power sharing of the disturbed PACES, while the required regulation response is maintained. The TESS rebound experiment further shows that BESS can compensate for the TESS power rebound.

- Extended validation under a representative 40-node campus topology, 4-hour RTDS-based HIL validation under time-varying regulation targets, and communication-robustness experiments under random delay and packet loss further support the applicability of the proposed method under larger PACES groups, longer operating horizons, and non-ideal communication conditions. In the proposed neighbor-to-neighbor architecture, the communication burden of each PACES is mainly determined by its local communication degree rather than the total PACES scale.
- The sensitivity analysis under five typical Chinese building climate zones shows that climate zone diversity affects the regulation sharing between BESS and TESS. Therefore, building climate zone characteristics should be considered in PACES coordination and configuration.

Future work will further investigate multi-season climate conditions, more stochastic PV fluctuation profiles, and practical engineering constraints in field deployment.

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